ChatGPT

Vaccine dose and regimen selection: current practice and the data needed for model-informed development

Bottom line

Dose and schedule selection remains predominantly **empirical, sequential and programme-specific**. WHO guidance envisages early trials exploring antigen amount, identifying a dose–response plateau, and then examining additional doses at different intervals; it does not prescribe a fixed number of dose levels or a formal dose-selection method. The current EMA vaccine guideline similarly treats dose, schedule, immunogenicity, safety and efficacy as components of the overall clinical development programme rather than specifying a pharmacometric framework. [1]

In prominent modern programmes, **roughly two to four active dose levels in an early trial is a reasonable description of practice**, often with only 10–20 participants per dose/age stratum initially. Three levels were used in the first mRNA-1273 trial; Pfizer/BioNTech tested 10, 20 and 30 μg of BNT162b2, alongside a 100 μg level for BNT162b1; Moderna tested four levels of its CMV vaccine; Janssen eventually studied four Ad26.COV2.S levels. These examples are not an industry-wide census, and programmes vary considerably. [2] A published review of the approved COVID-19 vaccines reached the broader conclusion that their early dose-ranging studies often left substantial uncertainty about whether the licensed dose was optimal. [3]

The important distinction for modelling is that **access to historical data is only part of the constraint**. Individual-level internal data from completed and discontinued programmes could be much more informative than publications. But a large archive of trials containing two or three doses at one interval does not reveal the joint dose–interval response surface. Likewise, heterogeneous assays, formulations and populations make absolute dose poorly transferable between vaccines. The most useful shared resource would therefore be longitudinal individual-level data tied to unusually detailed product, regimen and assay metadata, preferably from experiments containing genuine dose and interval contrasts.

How doses and schedules are selected in practice

Early human dose selection normally combines prior platform experience, non-clinical information, tolerability and one or more immunogenicity endpoints. The subsequent choice is commonly a judgement over whether higher doses add enough immune response to justify additional reactogenicity, rather than estimation of a well-characterised continuous dose–response curve. WHO explicitly describes preliminary trials as potentially testing different antigen amounts to describe dose–response and locate a plateau, followed by studies of subsequent doses at various intervals. [4]

Recent programmes illustrate the scale of the exercise:

Programme	Early exploration	What was carried forward
Moderna mRNA-1273	25, 100 and 250 µg; 15 adults per level; two doses 28 days apart	100 µg adult primary series. The trial varied dose, not interval. [5]
Pfizer/BioNTech BNT162	BNT162b2: 10, 20 and 30 µg; fixed 21-day interval; two related vaccine constructs were also compared	30 µg BNT162b2, again on the 21-day schedule. [6]
Moderna mRNA-1647 CMV	30, 90, 180 and 300 µg, all on a 0-, 2- and 6-month schedule	Later development narrowed the range; the phase I design was rich in dose but not schedule. [7]
Janssen Ad26.COV2.S	A later phase IIa study crossed four doses, one versus two vaccinations, and 56- versus 84-day intervals	An unusually informative example: the 84-day interval produced higher antibody responses than 56 days. [8]

Thus, **0–28 versus 0–56 is not ordinarily treated in the same way as two dose levels in a classical dose-ranging experiment**. Frequently an interval is fixed early and inherited by later trials. Pfizer used 21 days and Moderna 28 days throughout their pivotal adult COVID programmes; mRNA-1647 fixed 0, 2 and 6 months. [9] Janssen's explicit comparison of 56 and 84 days is closer to what a regimen-response study looks like. In the ChAdOx1 programme, the apparent benefit of longer prime–boost intervals emerged largely from differences between cohorts and trials rather than from a clean factorial interval experiment; pooled analyses subsequently found substantially greater immunogenicity and efficacy at longer intervals. [10] This is useful evidence about interval, but weaker for estimating an interval-response function than prospective randomisation across several intervals.

Schedule choice also contains considerations that have little to do with maximising a laboratory endpoint. WHO guidance explicitly treats age, maternal antibody, pre-existing immunity and compatibility with existing immunisation schedules as relevant to regimen selection. [11] That accords with the supplied Gates Foundation interview notes: for infant vaccines, a schedule that fits established 6–10–14-week or other routine visits may be more relevant operationally than the interval giving the highest titre.

What currently counts as leading model-informed practice

The clearest published example I found of a model actually informing a prospective clinical vaccine dose decision is Moderna's paediatric mRNA-1273 programme. An immunostimulatory/immunodynamic model pooled neutralising-antibody data across COVE, TeenCOVE and KidCOVE, incorporated age and dose, and predicted responses to dose–age combinations not directly studied. It projected that **25 µg would satisfy the prespecified immunogenicity criterion in children aged 2–5 years and infants aged 6–23 months; the subsequently observed trial results agreed with the prospective predictions**. The authors describe this as, to their knowledge, the first IS/ID model used in this way during clinical vaccine development. [12]

That is materially different from fitting a curve after a phase I study. It borrows information across age groups and dose levels, models the longitudinal immune process, propagates variability, and asks what would happen in an unobserved dose–population combination. It represents a plausible vaccine analogue of model-informed drug development. [13]

There is also a more advanced **research frontier** than routine industrial practice. A quantitative-systems-pharmacology model has been calibrated to longitudinal mRNA-1273 data and used to

simulate untested booster intervals and BNT162b2 doses; a newer multiscale mRNA model was calibrated to BNT162b2 and then adapted to mRNA-1273. These studies demonstrate transfer of common immune-system components across vaccines, but they are modelling demonstrations rather than evidence that such analyses routinely determine registrational doses. [14] Public evidence therefore suggests a substantial difference between leading practice at firms with mature platforms and modelling groups, and ordinary practice: **formal prospective model-based dose/regimen selection is possible, but still exceptional**. Moderna's 2025 paper itself states that pharmacometric approaches had not commonly been employed for vaccines. [13]

Which historical data would actually be informative

For modelling, the value of historical data is highly uneven.

Within the same platform or closely related vaccines, borrowing can be substantial. Common B-cell, plasma-cell and antibody kinetics can be estimated across studies, while vaccine-specific parameters govern stimulus magnitude. QSP work has explicitly reused common humoral-immunity mechanisms between products, and the 2025 mRNA model could be recalibrated from Pfizer's to Moderna's platform. [14] Within adenoviral vaccines, Benest and colleagues found evidence that host species and type of immune response predicted whether published dose–response curves were saturating or peaked, showing that at least *curve shape* can contain transferable information. [15]

Absolute optimal dose transfers much less readily. A microgram of mRNA, a microgram of recombinant protein and 10^{10} viral particles are not commensurable exposures; even within a platform, antigen sequence, expression, formulation, adjuvant, pre-existing immunity and manufacturing attributes can alter effective stimulus. The fact that host species predicted curve shape in the adenovirus analysis is itself evidence that unqualified pooling of animal and human doses is unsafe. [15]

A genuinely useful shared dataset would consequently be more than trial-arm GMTs:

Data layer	Information needed for modelling
Product	Platform; antigen identity and quantity; adjuvant and amount; vector/LNP or formulation characteristics; route and injection volume; lot, potency and relevant manufacturing/process version.
Regimen	Exact dose actually administered at every vaccination; exact dates rather than nominal “day 28”; number and order of doses; deviations; co-administered vaccines.
Participant	Age; baseline serostatus and quantitative immunity; previous infection/ vaccination; relevant immunosuppression/comorbidity; maternal antibody for infant studies; variables defining important exposure or population differences.
Longitudinal immunology	Individual, not merely arm-level, binding and functional antibody measurements; relevant B-cell and T-cell measures; samples around the post-dose rise/peak and during waning, extending sufficiently far to identify durability.

Data layer	Information needed for modelling
Assay metadata	Assay method, laboratory, strain/variant, units, calibration standard, limits of quantification and assay version, including bridging data when assays change. WHO specifically stresses standardisation and comparability of immunogenicity assays. [16]
Safety and outcomes	Individual solicited reactogenicity and safety data linked to each dose; infections or disease endpoints where available; censoring and follow-up. CDISC already defines standard representations for vaccine administration and reactogenicity, although its vaccine standard does not specify what experiments must be done. [17]
Preclinical linkage	Species, age, construct/formulation, absolute dose and route, schedule, longitudinal immunology and challenge outcomes, with enough metadata to distinguish genuine biological differences from assay differences.

The supplied interview observation that much useful material is unpublished is therefore important, particularly for **individual trajectories, low or failed doses, discontinued candidates and detailed assay/product metadata**. But access is not the only limiting factor. The Moderna IS/ID analysis worked because it could combine informative longitudinal measurements across several doses and age groups. [13] Conversely, no amount of data sharing can reconstruct a dose-by-interval interaction if virtually everybody received the same interval. This is the main distinction between an **access problem** and an **experimental-design problem**.

What such data could realistically predict for a new vaccine

The strongest use case is **conditional prediction after some human data from the new vaccine exist**. Historical data can provide priors for immune-system kinetics and variability; a small new-vaccine dataset can then anchor vaccine-specific potency. The resulting model could plausibly estimate the location and uncertainty of a plateau or peak, interpolate responses at untested doses, predict antibody time courses and waning, estimate effects of age or baseline immunity, and compare plausible booster intervals. Moderna's paediatric example establishes the feasibility of precisely this kind of interpolation across dose and age. [13] Platform QSP studies show the analogous possibility for interval and longer-term kinetic extrapolation. [14]

For a **completely new antigen or platform before human data**, the realistic target is narrower: a distribution over plausible curve shapes, kinetic parameters and informative dose ranges rather than a reliable numerical “optimal dose”. Cross-vaccine evidence such as the adenoviral analysis can sharpen those priors, but does not identify an absolute human dose. [15]

Finally, immunogenicity modelling does not by itself solve the efficacy problem. Where a validated or strongly supported immune correlate exists, predicted immune responses can potentially be translated into protection; where it does not, the mapping remains an additional uncertain model. Recent model-based meta-analysis in RSV illustrates that estimating an immunogenicity–efficacy relationship is a separate exercise requiring efficacy trials as well as immune data. [18] Rare safety outcomes are even less amenable to prediction from small dose-ranging datasets.

The practical implication for interpreting the historical-data question is therefore quite specific: **internal pharma data are potentially very valuable for learning transferable immune kinetics, between-person variability, platform effects and dose-response priors; they are much less capable of**

recovering schedule effects or precise optima where the original trials did not create those contrasts. The limiting factor is consequently access for some modelling tasks, assay and biological comparability for others, and—especially for joint dose/schedule optimisation—the sparsity of the experimental designs themselves.

---

1 4 11 16 cdn.who.int
https://cdn.who.int/media/docs/default-source/prequal/vaccines/who-trs-1004-web-annex-9.pdf?download=true&sfvrsn=9c8f4704_2

2 5 9 An mRNA Vaccine against SARS-CoV-2 — Preliminary Report | New England Journal of Medicine
https://www.nejm.org/doi/abs/10.1056/NEJMoa2022483?utm_source=chatgpt.com

3 Licenced doses of approved COVID-19 vaccines may not be optimal: A review of the early-phase, dose-finding trials.
https://pubmed.ncbi.nlm.nih.gov/37355453/?utm_source=chatgpt.com

6 Safety and Immunogenicity of Two RNA-Based Covid-19 Vaccine Candidates | New England Journal of Medicine
https://www.nejm.org/doi/abs/10.1056/NEJMoa2027906?utm_source=chatgpt.com

7 Safety and Immunogenicity of a Messenger RNA-Based Cytomegalovirus Vaccine in Healthy Adults: Results From a Phase 1 Randomized Clinical Trial.
https://pubmed.ncbi.nlm.nih.gov/38478705/?utm_source=chatgpt.com

8 Safety and immunogenicity of Ad26.COV2.S in adults: A randomised, double-blind, placebo-controlled Phase 2a dose-finding study - PubMed
https://pubmed.ncbi.nlm.nih.gov/38705804/?utm_source=chatgpt.com

10 Single-dose administration and the influence of the timing of the booster dose on immunogenicity and efficacy of ChAdOx1 nCoV-19 (AZD1222) vaccine: a pooled analysis of four randomised trials - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC7894131/?utm_source=chatgpt.com

12 13 Immunostimulatory/Immunodynamic model of mRNA-1273 to guide pediatric vaccine dose selection - Ivaturi - 2025 - CPT: Pharmacometrics & Systems Pharmacology - Wiley Online Library
https://ascpt.onlinelibrary.wiley.com/doi/10.1002/psp4.13237?utm_source=chatgpt.com

14 Application of quantitative systems pharmacology to guide the optimal dosing of COVID-19 vaccines - Giorgi - 2021 - CPT: Pharmacometrics & Systems Pharmacology - Wiley Online Library
https://ascpt.onlinelibrary.wiley.com/doi/10.1002/psp4.12700?utm_source=chatgpt.com

15 Response Type and Host Species may be Sufficient to Predict Dose-Response Curve Shape for Adenoviral Vector Vaccines - PubMed
https://pubmed.ncbi.nlm.nih.gov/32235634/?utm_source=chatgpt.com

17 Vaccines Therapeutic Area User Guide v1.1 | CDISC
https://www.cdisc.org/standards/therapeutic-areas/vaccines/vaccines-therapeutic-area-user-guide-v1-1?utm_source=chatgpt.com

18 Establishing Immune Correlates of Protection Against Respiratory Syncytial Virus Infection to Accelerate Vaccine Development: A Model-Based Meta-Analysis - Kashihara - 2026 - CPT: Pharmacometrics & Systems Pharmacology - Wiley Online Library
https://ascpt.onlinelibrary.wiley.com/doi/10.1002/psp4.70133?utm_source=chatgpt.com